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International Academic Research Competition 2026

Research Proposal

On

AI-Based Cancer Risk Prediction in the Population of Bangladesh

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Grade: A-Level

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Research Problem:

Cancer is a major public health concern in Bangladesh, with an estimate of 167,256 new cases and 116,598 cancer deaths in 2022 [1]. Cancer risk is influenced by genetic factors, age, family history, lifestyle and environmental exposures [2,3]. In Bangladesh, important risk factors include arsenic-contaminated drinking water, tobacco use, betel quid and areca-nut chewing, unhealthy diet and air pollution [3–5]. However, these risk factors are often studied separately, although several of them may affect the same person at the same time.

A more integrated approach to assess the risk of cancer is needed in the population of Bangladesh. Artificial Intelligence (AI) can analyse large amounts of information and identify patterns among multiple risk factors [6]. However, limited research has explored the combined use of genetic, environmental and lifestyle factors for AI-based cancer-risk prediction in Bangladesh. This project aims to explore whether combining these factors can improve cancer risk assessment and identify individuals who may benefit from earlier or more targeted screening.

Research Questions:

This study aims to answer the following questions:

1. Can an AI-based approach combining genetic, environmental and lifestyle factors predict cancer risk among adults more accurately than any single-factor based risk assessment in Bangladesh?

2. Which genetic, environmental and lifestyle factors contribute most significantly to cancer risk and which combinations of genetic susceptibility with exposures such as arsenic, tobacco use and other lifestyle factors are associated with increased risk of cancer?

3. Can AI-based cancer-risk assessment identify individuals with higher predicted risk and estimate how changes in variable factors, such as tobacco use and household smoke exposure may reduce their predicted risk and support earlier or more targeted screening?

Existing Literature:

Previous studies have examined genetic, environmental, lifestyle and AI-based approaches to cancer-risk assessment. Yang et al. [2] and Collatuzzo and Boffetta [3] describe genetic changes, tobacco use, diet, obesity and environmental exposure as significant factors affecting the risk of cancer. Das [4] reported an association between arsenic exposure from drinking water and cancer risk in Bangladesh. Betel quid and areca-nut chewing was identified as important cancer risk factors by Jasim et al. [5]. Byrne et al. [7] showed that Polygenic Risk Scores (PRS) can help estimate inherited cancer susceptibility. These studies provide strong evidence that cancer risk is influenced by multiple genetic, environmental and lifestyle factors.

More recently, Ahmed et al. [6] reviewed the use of machine for cancer-risk prediction and highlighted the need for better justification in different populations. Moreover, O'Donnell et al. [9] demonstrated the use of machine learning techniques for predicting time-to-event outcomes, while Kachuri et al. [8] showed that the performance of genetic risk scores can vary across populations. These studies support the use of AI and genetic risk assessment for predicting cancer risk.

AI-based approaches can analyse several risk factors simultaneously and recognise patterns that may be difficult to detect using traditional methods. This is especially important in cancer, as a person's genetic susceptibility can interact with environmental and lifestyle factors. For example, genetic risk can be assessed alongside tobacco use, arsenic exposure, diet and other modifiable factors to provide a more personalised estimate of an individual's cancer risk [3–7].

However, most existing studies have focused on individual risk factors, specific cancer types or populations outside Bangladesh. There is limited research combining genetic, environmental and lifestyle factors in a single AI-based cancer-risk prediction approach for the Bangladeshi population. This study aims to address this gap by combining these factors to estimate an individual's cancer risk and identify those who may benefit from earlier or more targeted screening.

Methodology:

1. Study Design: This study will use a prospective cohort design. At the beginning of the study, information on health, lifestyle, environmental exposure, and selected genetic factors will be collected from adults from both rural and urban areas of Bangladesh. Participants will then be followed through health and hospital records to identify new cancer diagnoses [6,9].

2. Study Population and Sample: Approximately 10,000 men and women aged 30–65 years will be recruited from selected rural and urban districts of Bangladesh. Participants with a previous cancer diagnosis will be excluded from the risk-prediction cohort.

3. Data Collection:

I. Quantitative Data: Blood samples will be collected to assess selected genetic variants and calculate genetic risk scores. Household drinking-water samples will be tested for arsenic concentration. Height, weight, and blood pressure will also be recorded. Local air-pollution exposure will be estimated using available monitoring data [3,4].

II. Qualitative Data: Trained health workers will conduct structured face-to-face interviews in Bengali. Information will be collected on diet, physical activity, indoor cooking-smoke exposure, smoking, and use of smokeless tobacco such as zarda and gul [3,5]. These responses will be converted into quantitative variables or scores for analysis.

4. Data Analysis:

I. Dependent Variable: The main dependent variable will be the time from enrolment to a verified new cancer diagnosis or the end of follow-up. Participants who do not develop cancer during follow-up will be treated as censored observations in the survival analysis [9].

II. Independent Variables: The main independent variables will include genetic risk scores, arsenic exposure, air pollution, indoor cooking-smoke exposure, tobacco use, diet, physical activity, age and other relevant factors [3–5,7].

III. Statistical and AI-Based Analysis: Data will be cleaned analysed using Python or R. An AI-based survival-analysis approach will be used to estimate cancer risk. DeepSurv will be used for survival prediction and compared with Random Survival Forests (RSF) [9,10]. Prediction performance will be assessed using the Concordance Index (C-index) to determine how well the approach distinguishes participants according to their observed time to cancer development [11].

6. Ethical Considerations: Written informed consent will be obtained from all participants in Bengali. Personal identifiers will be replaced by study IDs and participant information will be kept confidential. Individual research findings will be handled according to the approved ethical protocol.

Expected Outcome:

We expect the AI-based approach to provide a more comprehensive assessment of the risk of developing cancer. The study will identify individuals with higher predicted risk and determine which genetic, environmental and lifestyle factors contribute most strongly to their risk.

The analysis will also examine whether arsenic exposure and genetic susceptibility are associated with cancer risk [4,8]. SHAP analysis will help explain how individual factors contribute to each participant's predicted risk [11]. The study will therefore assess whether integrating genetic, environmental and lifestyle information can improve cancer-risk assessment in the Bangladeshi population and support the identification of individuals who may benefit from earlier or more targeted screening.

Potential Limitations:

The study may be limited participant dropout, incomplete health records, and differences in the quality of genetic, environmental and lifestyle data. The findings may also require validation in larger and more diverse Bangladeshi populations before the approach can be used in routine cancer-risk assessment.

Conclusion:

16 Cancer is a major public health burden in Bangladesh, where genetic, environmental, and lifestyle factors may interact to influence an individual's risk of developing cancer. This study will assess whether AI can combine these factors to improve cancer-risk prediction, identify key contributors to risk and support earlier or more targeted screening. If validated, the approach could provide a practical, personalised and locally relevant tool for cancer-risk assessment and prevention across rural and urban parts of Bangladesh.

Project Implementation Plan:

The project will run for approximately 5 years, beginning with research planning, literature review, ethical approval and preparation of data-collection tools during the first 6 months. Next 4 years will focus on participant recruitment, collection and cleaning of genetic, lifestyle, and environmental data, development of genetic risk scores and initial AI-based risk analysis. The last 6 months will involve prediction analysis, validation, interpretation of risk factors, preparation of figures and tables and completion of the final research report.

Future Directions:

The researchers will further validate and improve the AI-based approach, share the findings through academic publications and conferences, explore its wider use in Bangladesh and collaborate with government and public health organisations to develop a practical, cost-effective tool that can support cancer prevention, early screening and better health outcomes for the people of Bangladesh.

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